Long-Tail and Open-Set Recognition of Brain Tumours with Evidence-Grounded Reports

A feasibility pilot with the Virchow pathology foundation model on the EBRAINS Digital Brain Tumour Atlas

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Why rare brain tumours

- The WHO CNS5 classification lists more than 100 tumour types; most are rare.
- In the EBRAINS atlas, 30 common types hold 2,298 slides, while about 96 rare types share about 770.
- Published work on this atlas classifies the common types only.
- A model in practice will also meet types it has never seen, and it should say so.

Questions for this pilot

1. Do frozen Virchow features separate common *and* rare types?
2. Can we flag a slide whose type was never seen?
3. Can we show which tissue drove each call?
4. Can a report stay grounded in that evidence?

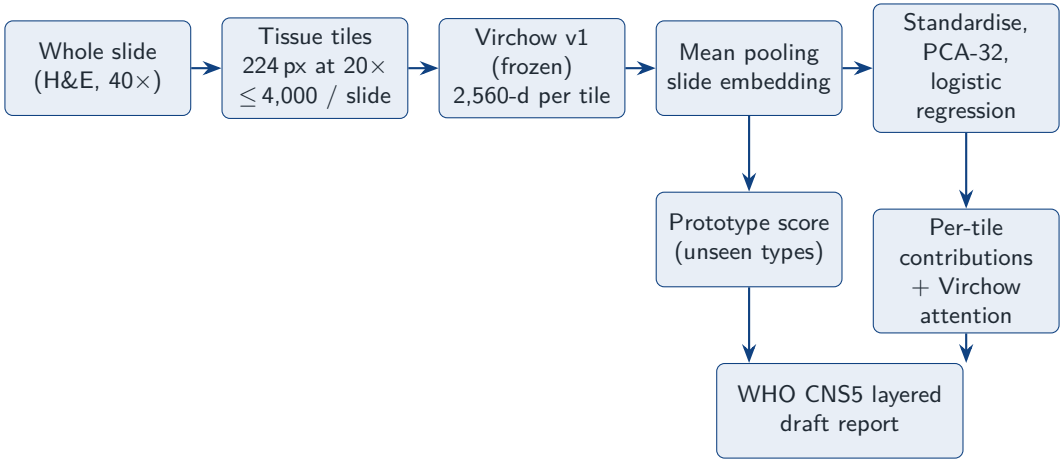
Data: a 42-slide pilot from the EBRAINS atlas

- EBRAINS Digital Brain Tumour Atlas: 3,115 H&E whole-slide images, 126 types, 3.6 TB.
- Pilot chosen by type frequency, one slide per patient where possible.
- Each slide (about 1.2 GB, `.ndpi`) is downloaded, encoded, then deleted.

Research use only: Virchow and PRISM licences forbid clinical diagnosis.

Role	Tumour type	Slides
Common	Glioblastoma, IDH-wt	5
	Meningothelial meningioma	5
	Pituitary adenoma	5
	Pilocytic astrocytoma	5
Rare	CNS lymphoma	4
	Leiomyosarcoma	4
	Immature teratoma	4
	Ependymoma, RELA-fusion	4
Held out	Haemangioblastoma	3
	Anaplastic astrocytoma, IDH-wt	3

Pipeline



Feature extraction on a free Kaggle GPU (about 3–4 min per slide); everything after it runs on a laptop CPU.

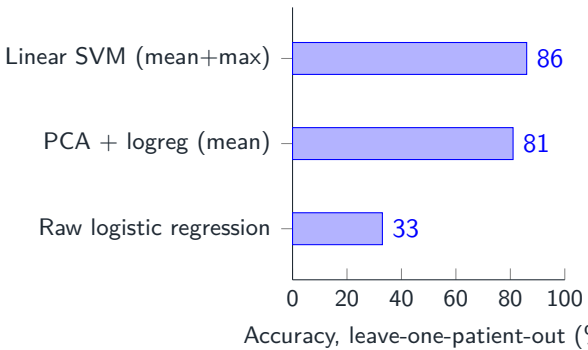
Evaluation protocol

- **Known types** (36 slides): leave-one-patient-out. The classifier is refit without the test slide and without any other slide of the same patient.
- **Unseen types**: each rare or held-out type is hidden from training in turn (6 rotations). The detection threshold is set on known slides only, at 95% of them accepted.
- **Classifier choice**: 12 configurations compared (logistic regression with and without standardisation, PCA, k-NN, nearest centroid, linear SVM; mean or mean+max pooling).

Caveat

42 slides is a feasibility check. Every number below has a wide confidence interval and must be confirmed on new slides.

Result 1: Virchow features separate common and rare types



- Standardising the features was the key step: 33% → 81%.
- We keep PCA + logistic regression: calibrated probabilities, and linear, so its score splits exactly over tiles.
- Per slide: **29/36** correct; common 18/20, rare 11/16.
- Hardest: CNS lymphoma (1/4), confused with glioblastoma.

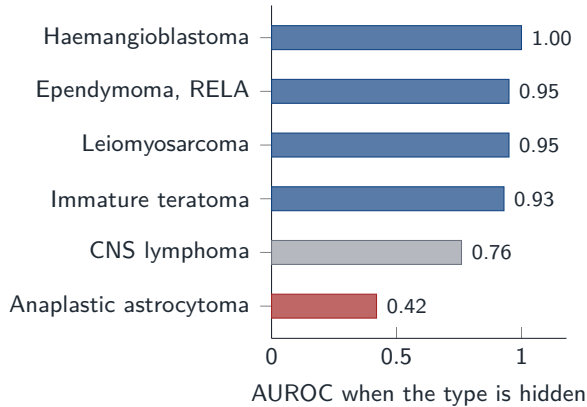
86% is the best of 12 configurations, hence optimistic.

Result 2: flagging tumour types never seen in training

Prototype score: distance from the slide embedding to the closest known-type mean, after per-dimension standardisation.

	L2 only	Standardised
AUROC	0.63	**0.83** $\pm$ 0.20
Unseen caught	18%	**50%**
Known falsely flagged	–	2.6%

Mean over 6 leave-one-type-out rotations; threshold from known slides only.



Far versus near out-of-distribution

Far OOD: well detected

Haemangioblastoma, RELA ependymoma, leiomyosarcoma and immature teratoma look unlike every known type (AUROC ≥ 0.93).

Intermediate

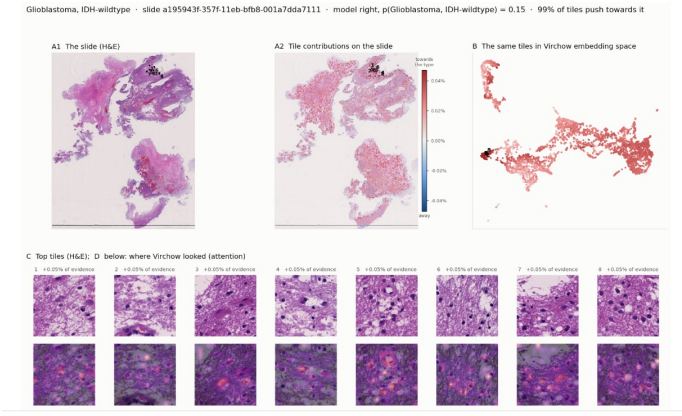
CNS lymphoma (0.76): partly overlaps glioblastoma, as in the closed-set confusions.

Near OOD: not detected

Anaplastic astrocytoma, IDH-wt (0.42) sits inside the glioblastoma and pilocytic astrocytoma cloud. None of the five scores we tested (confidence, energy, Mahalanobis, k-NN, prototype) separates it.

Consequence: report far and near OOD separately; a hierarchical level ("glioma, subtype uncertain") is a natural next step.

Result 3: which tiles made the call



Mean of tile embeddings + linear classifier $\Rightarrow$ the score splits exactly over the n tiles, $s_c = \sum_i \frac{1}{n}(w_c - \bar{w})^\top t_i + b_c$: each term is that tile's exact Shapley value. A1 slide, A2 contribution map, B tiles in Virchow space, C top H&E tiles, D Virchow attention.

Glioblastoma slide; figure from an earlier, unstandardised classifier, hence the low probability.

Baseline: PRISM reports from the same Virchow features

Tumour type	Type named correctly
Pituitary adenoma	5/5
Pilocytic astrocytoma	4/5
CNS lymphoma	4/4
Leiomyosarcoma	4/4 (as sarcoma)
Glioblastoma, IDH-wt	2/5
Haemangioblastoma	2/3
Ependymoma, RELA	1/4
Meningothelial meningioma	0/5
Immature teratoma	0/4
Anaplastic astrocytoma	0/3
Total	**about 22/42**

- Good on visually distinctive types; never names meningioma.
- Reaches for diagnoses from outside the brain (sarcoma, GIST, carcinoma).
- Makes up findings: wrong organ (lymph node, uterus), immunostains and Ki-67 that H&E cannot show.
- One fluent line, not a structured report.

Graded by the authors, not yet by a neuropathologist.

Result 4: evidence-grounded WHO CNS5 draft reports

Layered format of the ICCR dataset, every field filled from facts the pipeline can show:

- **Integrated diagnosis**: type, given as NOS because molecular status is unknown
- **Histopathological classification** and **CNS WHO grade**
- **Molecular information**: "not assessed", with the tests to request
- **Microscopic findings**: probability, tiles analysed
- **Comment**: differential and coordinates of the regions to review

Safety rules

- Below $p = 0.5$: "Indeterminate on H&E" with a differential.
- Above the prototype threshold: "tumour type not recognised by the model", with methylation profiling suggested.
- An optional LLM may rephrase findings, but its text is rejected if it contradicts the diagnosis.

Every report is marked as a research draft.

Limitations

- **Small sample**: 42 slides, 3–5 per type; confidence intervals are wide.
- **Optimistic selection**: the classifier was chosen on the same slides it was scored on.
- **Ten types only**, out of 126 in the atlas.
- **No expert review yet**: PRISM grading and the WHO mapping table need a neuropathologist.
- **H&E only**: IDH, fusions and methylation class, central to WHO CNS5, are out of reach.
- **Single magnification** ($20\times$), at most 4,000 tiles per slide.

Next steps

Data and models

- 15–20 slides per type, more rare types
- Attention-based MIL pooling (ABMIL)
- Multi-magnification features
- Hierarchical labels for near-OOD gliomas

Evaluation

- Confirm accuracy and AUROC on new, untouched slides
- Report far and near OOD separately
- Neuropathologist rating of reports: correctness and made-up findings, ours vs PRISM

Take-aways

1. Frozen Virchow v1 features, standardised, recognise common *and* rare brain tumour types in a 42-slide pilot (29/36 slides, leave-one-patient-out).
2. A standardised prototype score flags clearly different unseen types (AUROC $\geq$ 0.93) but not near-OOD gliomas.
3. A linear classifier on mean-pooled tiles gives exact, cheap per-tile explanations.
4. Structured WHO CNS5 reports state only what this evidence supports; PRISM, on the same features, makes up organs and stains. Expert rating is the next test.

Code: `github.com/AsmaActiveLearning/Cancer_diagnosis`

Thank you

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Data: EBRAINS Digital Brain Tumour Atlas (Roetzer-Pejrimovsky et al., 2022).
Models: Virchow v1 and PRISM (Paige), research use only.